Attacking and hardening models

AI security · University · Lesson 3



In this lesson you will

- Explain evasion, extraction and membership inference.
- Compute the worst small nudge for a linear model.
- Choose defences and measure what they cost.

Remember

- Confidence isn't robustness.
- For a linear score, the worst nudge moves each input ε against its weight; the score drops by ε × the sum of $|w|$.
- Every defence costs accuracy, usefulness or convenience: measure it.

Nudge the score by hand

1. Warm-up (3 min): Is 99% sure the same as safe?
2. Score (6 min): Weights 2, -1, 1; bias -1; every input 1.
3. Nudge (6 min): Move each input 0.5 against its weight; score again.
4. Certify (5 min): How small must ε be so the answer can't flip?

Work on paper: the numbers are small on purpose.

Let’s discuss

1. Why doesn't a confident answer mean a robust one?
2. Which defence would you pay for, and what would it cost?
3. Why do exact scores help an attacker?

Check yourself

- Weights 3 and -2, inputs 1 and 1, bias 0, ε = 0.5: write the worst nudge and the new score.